



Speed, Distance, Time **Answers**

Unless otherwise stated, give your answer to any speed questions in kilometres per hour (km/h).

1. Calculate the average speed of a car travelling 100km in 2 hours.

$$\frac{100}{2} = 50\text{km/h}$$

2. Calculate the average speed of a rambler who travels 12km in 3 hours.

$$\frac{12}{3} = 4\text{km/h}$$

3. Calculate the average speed of a train travelling 25km in 20 minutes.

$$20 \text{ mins} = \frac{1}{3} \text{ of an hour} \quad 25 \div \frac{1}{3} = 75\text{km/h}$$

4. Calculate the distance travelled by a cyclist travelling at 7km/h for 5 hours.

$$7 \times 5 = 35\text{km}$$

5. Calculate the distance travelled by a swimmer swimming at 3km/h for 30 minutes.

$$30 \text{ minutes} = \frac{1}{2} \text{ an hour} \quad 3 \times \frac{1}{2} = 1.5\text{km}$$

6. Calculate how long it would take for a car to travel 50km if it was travelling at an average speed of 20km/h. Give your answer in minutes.

$$\frac{50}{20} = 2.5 \text{ hours} \quad 2.5 \text{ hours} = 150 \text{ minutes}$$

7. Calculate how long it would take a runner to run 5km if they were running at an average speed of 4km/h. Give your answer in hours and minutes.

$$\frac{5}{4} = 1.25 \text{ hours} \quad 1.25 \text{ hours} = 1 \text{ hour } 15 \text{ minutes}$$

8. Emma travels from Newcastle to London, a distance of 283 miles, by car. It takes her 4 hours and 20 minutes. Pete does the same journey by train. The train travels 285 miles at an average speed of 90mph.

- a. Assuming Emma and Pete leave at the same time, who will arrive in London first?

$$\text{time} = \text{distance} \div \text{speed}$$

$$= 285 \div 90$$

$$= 3.166... \text{ hours}$$

$$= 3 \text{ hours } 10 \text{ minutes}$$

Pete will arrive first.

- b. Explain why Pete's speed is an average speed.

The train will stop at stations so the speed will not be constant.

- c. Find Emma's average speed in miles per hour. Give your answer correct to 1 decimal place.

4 hours and 20 minutes = 4.33... hours

speed = distance ÷ time

$$= 283 \div 4.33...$$

$$= 65.3\text{mph (1d.p.)}$$

9. Sofia can run 200 metres in 26 seconds.

- a. Estimate the time it will take her to run 300m, giving your answer in seconds.

speed = distance ÷ time

$$= 200 \div 26$$

$$= 7.69... \text{ seconds}$$

time = distance ÷ speed

$$= 300 \div 7.69...$$

$$= 39 \text{ seconds}$$

- b. Sofia says, "It will take me 3 minutes 15 seconds to run 1500m". Do you agree with Sofia?

$$1500 \div 300 = 5$$

$$39 \times 5 = 195 \text{ seconds}$$

$$= 3 \text{ minutes and } 15 \text{ seconds}$$

Mathematically, Sofia is correct. However, physically, it is unlikely that she would be able to maintain the same speed over a much longer distance.

Challenge

Mira is sailing from Liverpool, England to Liverpool, Canada. They want to beat their personal record of 43 days, 4 hours and 23 minutes. The total distance is 2792 miles.

Unfortunately, Mira has trouble with their boat and has to stop the voyage at St John's, Canada. Mira has been sailing for 31 days, 8 hours and 34 minutes and has travelled a distance of 2175 miles.

Was Mira on track to beat their personal record?

Liverpool, England → Liverpool, Canada

$$43 \text{ days} = 43 \times 24 = 1032 \text{ hours}$$

$$23 \text{ minutes} = 23 \div 60$$

$$= 0.383... \text{ hours}$$

$$\text{Total time} = 1032 + 4 + 0.383...$$

$$= 1036.383... \text{ hours}$$

$$\text{Speed} = 2792 \div 1036.383... = 2.69\text{mph (2d.p.)}$$

Liverpool, England → St John's, Canada

$$31 \text{ days} = 31 \times 24 = 744 \text{ hours}$$

$$34 \text{ minutes} = 34 \div 60$$

$$= 0.566.... \text{ hours}$$

$$\text{Total time} = 744 + 8 + 0.566...$$

$$= 752.566... \text{ hours}$$

$$\text{Speed} = 2175 \div 752.566... = 2.89\text{mph (2d.p.)}$$

Yes, Mira was on track to beat their previous record.